

Yuxuan Shu

PHD CANDIDATE · COMPUTER SCIENCE, CENTRE FOR ARTIFICIAL INTELLIGENCE

University College London, London, WC1E 6BT, United Kingdom

✉ yuxuan.shu.22@ucl.ac.uk | 🏠 claudiashu.github.io | 📧 ClaudiaShu | 📞 yuxuan-shu-015137198

Education

University College London

London, United Kingdom

PHD IN COMPUTER SCIENCE, CENTRE FOR ARTIFICIAL INTELLIGENCE

Dec. 2022 – present

- Supervisor: Dr. Vasileios Lamos
- Published 2 peer-reviewed papers at top-tier venues (AAAI oral, TMLR) and 2 preprints
- Developed novel architectures for time series models including deformable attention mechanisms and spectral operators

Imperial College London

London, United Kingdom

MASTER OF RESEARCH, MEDICAL ROBOTICS AND IMAGE GUIDED INTERVENTION

Oct. 2021 – Oct. 2022

- Supervisor: Dr. Benny Lo
- Graduated with distinction
- Published 1 peer-reviewed paper at a top-tier venue (BMVC)

Wuhan University

Wuhan, China

BACHELOR, REMOTE SENSING SCIENCE AND TECHNOLOGY

Sept. 2017 – Jun. 2021

- Undergrad research advisor: Dr. Fan Zhang

Internships

Nokia Bell Labs

Cambridge, UK

RESEARCH INTERN

Jun. – Aug. 2025

- Supervised by Dr. Mohammad Malekzadeh and Dr. Peter Charlton
- 1 preprint paper; 1 patent filed; code open-sourced

Publications

PUBLISHED

Yuxuan Shu and Vasileios Lamos. Sonnet: Spectral Operator Neural Network for Multivariable Time Series Forecasting, **Association for the Advancement of Artificial Intelligence (AAAI)**, 2026 (**Oral Presentation**) • Paper | Code | Dataset

Yuxuan Shu and Vasileios Lamos. DeformTime: Capturing Variable Dependencies with Deformable Attention for Time Series Forecasting, **Transactions on Machine Learning Research (TMLR)**, 2025 • Paper | Code | Project

Yuxuan Shu et al. Revisiting Self-Supervised Contrastive Learning for Facial Expression Recognition, **British Machine Vision Conference (BMVC)**, 2022 • Paper | Code | Project

PREPRINTS

Yuxuan Shu et al. CLEF: Clinically-Guided Contrastive Learning for Electrocardiogram Foundation Models, Preprint, 2025 • Paper | Code

Yuxuan Shu et al. Foundation models for biosignals: A survey, Preprint, 2025 • Paper | Code

Yuxuan Shu and Vasileios Lamos. Unsupervised hard Negative Augmentation for contrastive learning, Preprint, 2024 • Paper | Code

Presentations

INVITED TALKS

2026. *Foundation Models for Biosignal Analysis*. Invited Research Talk: Computational Health Informatics (CHI) Lab, University of Oxford, Oxford, UK.

2026. *Time Series Forecasting in the Era of Deep Learning*. Invited Lecture Talk: COMP0084, UCL, London, UK.

2025. *Capturing Variable Dependencies with Deformable Attention for Time Series Forecasting*. Invited Seminar Talk: Financial Computing & Analytics (FCA) Seminar, UCL, London, UK.

CONTRIBUTED PRESENTATIONS

Sonnet: Spectral Operator Neural Network for Multivariable Time Series Forecasting. AAAI 2026 Oral, Singapore.

Patents

Patent Filed

SYSTEM AND ALGORITHMS FOR PRE-TRAINING AND FINE-TUNING OF A SINGLE-LEAD ECG FOUNDATION MODEL WITH CLINICALLY GUIDED RISK SCORE OF FUTURE HEALTH OUTCOMES 2025

Collaborations

Dr Xiao Gu, University of Oxford. Joint research on foundation models and biosignal analysis.
Dr Mohammad Malekzadeh, Microsoft (formerly Nokia Bell Labs). Collaboration on ECG foundation models.

Honors & Scholarships

- 2022–2026 UCL CS Departmental Funds for PhD Studentship, University College London
- 2021–2022 Hamlyn Prize, Top 3 student, Imperial College London
- 2017–2018 Academic Excellency Scholarship, Top 5%; Wuhan University

Teaching Experience

COMP0084 Information Retrieval and Data Mining GRADUATE TEACHING ASSISTANT	University College London 2023–2024; 2024–2025
COMP0087 Statistical Natural Language Processing GRADUATE TEACHING ASSISTANT	University College London 2023–2024; 2024–2025
COMP0257 Advanced Topics in Financial Modeling and Technologies GRADUATE TEACHING ASSISTANT	University College London 2024–2025; 2025–2026

Outreach & Professional Development

- PEER REVIEW Reviewed 50+ papers at top-tier ML/AI conferences: NeurIPS, ICML, ICLR, AAAI, AISTATS, ACL, EMNLP.
- OPEN-SOURCE CONTRIBUTIONS Open-sourced all research implementations; projects have 150+ GitHub stars.

Technical Skills

- DEEP LEARNING Transformers, diffusion models, self-supervised and contrastive learning, multivariate time series modeling, uncertainty quantification, probabilistic forecasting.
- FRAMEWORKS PyTorch, PyTorch Lightning, TensorFlow, HuggingFace, scikit-learn, NumPy, Pandas.
- ENGINEERING Distributed training pipelines, large-scale data preprocessing, experiment tracking (Weights & Biases, Hydra), reproducible research practices.
- DEVOPS Docker, Git, CI/CD workflows, model evaluation and benchmarking.